KAHM Forecasting: Revision Review

Step 2 of the KAHM forecasting research plan

Prepared for Mohit Kumar 31 October 2026

1. Outcome and scientific judgment

The KAHM forecasting manuscript has been revised in its editable LaTeX sources. Its new title is:

> **Kernel Affine Hull Features for Time-Series Forecasting:**
> **Feature Spans, Bounded Corrections, and Matched Evaluation**

The subtitle replaces "Distinctness, Bounded Corrections, and Predictive Limits." It describes the main results directly and distinguishes this paper from the risk-certificate manuscript revised in Step 1 and the earlier prediction-limit paper [4,5].

The paper's strongest case is an analysis of when KAHM features add functions to a matched decoder and whether those features improve forecasting in the declared experiment. The evidence supports a useful feature contribution beyond the shared local-coordinate basis. It does not establish the prespecified forecasting advantage over the entire comparison family. That distinction is retained throughout the revision.

The work completed here includes a contribution and literature review, proof review, a repair to one supporting proof, explicit matching of the comparison scopes, fresh independent mathematical diagnostics, and independent arithmetic checks of the principal saved forecasts. No new forecasting experiment was run and no reported experimental value was changed.

What the revised manuscript establishes

1. **A precise comparison of raw feature spans.** Sufficient reference-symmetry conditions make the augmented KAHM and positive-kernel RBF spans equal. An explicit irregular reference bank gives a KAHM coordinate outside the specified RBF span. This is not a universal ordering of the two methods.
2. **A deterministic bound on the fitted correction.** A bounded feature map and centered ridge calculation control the residual head, including its free intercept. The result applies to other bounded features as well; it does not bound total forecast error.
3. **A reproducible, qualified empirical finding.** KAHM improves on the common local-coordinate basis under the declared nonlinear criterion and satisfies the linear noninferiority criterion. The remaining four nonlinear criteria are not established.

Delivered material

The revised main paper has **23 pages**; the self-contained *Supplementary Studies* has **71 pages**. The editable source archive includes both PDFs, this review and its source, a change patch, audit programs, their recorded outputs, and compact provenance records. The recovered original saved-forecast ZIP is delivered separately. The official TMLR style files and all 65 original generated table/figure assets are unchanged.

2. Applied revision log

Location	Applied change and purpose
Title and abstract	Uses “Feature Spans” and “Matched Evaluation” to state the central scope. The abstract now attributes normalized target transfer to an interpretation of the established KAHM construction and connects it to rescaled interpolation. All experimental numbers and the qualified conclusion remain intact.
Introduction	Credits the original KAHM construction and the rescaled interpolation antecedents. States the specific span results, intercept-inclusive correction bound, and forecasting evidence as the paper’s contributions.
Related work	Separates existing geometric reconstruction results from the present fitted residual-head bound. Describes the local-coordinate baseline as having the same gate/local-coordinate form, without claiming identical fitted features. Acknowledges the predecessor’s risk certificates, fitted upper bounds, and common-target comparisons.
Relation to Step 1	Adds an explicit comparison with *A Hierarchy of Risk Certificates for Forecasting Representations*, cited as an unpublished manuscript. Distinguishes class-level risk/cost certificates from the current raw-span and fitted-procedure results.
Normalization result	Explains that, when the Gram matrix is invertible and the denominator is nonzero, setting regularization to zero recovers the established rescaled kernel interpolant. Positive-regularization KAHM target transfer is presented as an interpretation, not a new normalization principle.
Span comparison	Makes explicit that the common span contains the gated affine reconstruction. Adding a departure therefore gives the same raw span as adding its corresponding gated nonlinear reconstruction. This clarifies the role of affine subtraction.
Experimental comparison	Separates matched KAHM/RBF reconstruction controls from complete-procedure comparators. States the 42 common plus 168 departure coordinates and contrasts them with the raw-RBF representation and its different search space.
Main mathematical appendix	Corrects three inconsistent occurrences of the query variable in the irregular-reference proof, including its Gaussian common factor. The argument and conclusion are unchanged.
Supporting mathematical appendix	Repairs global admissibility in the reconstruction-to-prediction bridge using an explicit scalar Lipschitz extension. No stronger hypothesis or altered conclusion is needed.
Build and reproduction notes	Identify the present local manuscript revision separately from the public experiment commit. Document the recovered numerical supplement, independent replay commands, expanded audit input requirements, and the author package’s identifying material.

Preserved evidence. The tables, figures, parameter grids, margins, confidence allocation, and original supporting-study results are retained. No unfavorable comparison, incomplete evaluation, or disclosed protocol limitation was removed. Title wrapping, a sentence interrupted by a table, and a detached subsection heading were also corrected. The source patch makes every textual edit inspectable.

3. Mathematical review and proof repair

Normalized target transfer and its antecedent

For $C = (K + \lambda I)^{-1}$ and $s(z) = \mathbf{1}^\top Ck(z) \neq 0$, the normalized weights satisfy

$$\omega(z)^\top y = \frac{k(z)^\top (K + \lambda I)^{-1} y}{k(z)^\top (K + \lambda I)^{-1} \mathbf{1}}.$$

The numerator and denominator are kernel-ridge predictions without an additional intercept. Symmetry of C establishes the identity. The revised attribution credits rescaled RBF interpolation [1,2] and the original normalized KAHM construction [3], avoiding novelty claims for the normalization alone.

What the span results do and do not imply

If $\mathcal{S}_0$ contains the common gated affine coordinates, then

$$\operatorname{span}\{\mathcal{S}_0, g(A_K - A_A)\} = \operatorname{span}\{\mathcal{S}_0, gA_K\},$$

and the same statement holds for A_R. The equal-span proposition uses constant Gram row sums and invariance of $\operatorname{span}\{\mathbf{1}, Q_{:1}, \ldots, Q_{:d}\}$ under C. The restriction of C to that invariant finite-dimensional space is onto; the auxiliary coordinate matrix B need not be invertible. Redundant reference-coordinate columns therefore do not invalidate the proof.

The irregular-bank example establishes a function outside the specified comparison span, not universal containment or distinctness for every fitted bank. Equal raw spans need not give identical ridge fits: coordinate changes alter the isotropic penalty, and separate nonlinear radial transformations can change span relations. Exact quotient statements do not automatically extend across a fallback boundary.

The correction bound includes the free intercept

Eliminating the intercept from the mean-squared-error ridge objective gives $b = \bar{e} - \bar{\psi}^\top w$. Its centered spectral calculation yields

$$\|w\|_2 \leq \frac{\sigma_e}{2\sqrt{\alpha}}, \qquad |\gamma q(x)| \leq |\gamma| \left[|\bar{e}| + (R + \|\bar{\psi}\|_2) \frac{\sigma_e}{2\sqrt{\alpha}} \right].$$

The conditions are $\alpha > 0$, $n_A \geq 1$, and $\|\psi(x)\|_2 \leq R$. The baseline and full history-to-feature sensitivity remain uncontrolled; the radial map is nonexpansive only in its own input.

Repair to the supporting reconstruction bridge

The supporting proposition assumes that the scalar conditional mean m is L-Lipschitz on a domain D containing X and $h(U)$ almost surely. The original argument used $m \circ h$ as a globally admissible decoder, although $h(u)$ need not belong to D at every representation point. The revised proof defines

$$\widetilde{m}(x) = \inf_{z \in D} \{m(z) + L\|x - z\|_2\}.$$

This finite, globally L-Lipschitz extension agrees with m on D. Then $g = \widetilde{m} \circ h$ is measurable and globally LL_h-Lipschitz, agrees with $m(h(U))$ almost surely, and retains the original square-integrability and risk bound. This closes the domain gap without changing the proposition's assumptions or conclusion.

4. Principal experimental finding retained

The principal study contains four fixed simulation laws, two observation noise conditions, and 24 repetitions: **192 task suites**. The independent statistical units are the **24 repetition aggregates**, not the individual forecast windows or all 192 task suites. Histories have length 12 and forecasts use horizons 1, 4, and 12. Evaluation D contains 1,664 origins per task. The protocol remains locally prospective, informed by supporting work, and not publicly preregistered.

Nonlinear comparisons

The bounded KAHM-departure procedure has mean normalized MSE **0.0299267**. Positive gain below means lower KAHM loss relative to the comparator; negative gain means higher KAHM loss.

Comparator	Mean NMSE	KAHM gain	Margin lower bound	Established?
Common local coordinates	0.0342342	12.58%	0.00295461	Yes
Bounded RBF departures	0.0302808	1.17%	-0.00078620	No
Raw RBF forecasting	0.0311463	3.92%	-0.00008286	No
Histogram gradient boosting	0.0294064	-1.77%	-0.00193630	No
C-selected non-KAHM policy	0.0282223	-6.04%	-0.00298995	No

The nonlinear margin contrasts are $0.97L_{\text{comparator}} - L_{\text{KAHM}}$. Their one-sided lower bounds use the original Bonferroni allocation over all six primary comparisons. The table is recomputed from the saved forecasts; its values agree with the reported analysis to floating-point tolerance [8].

For the linear control, KAHM mean NMSE is 0.898104 versus ridge 0.896320. The lower bound for the specified 5% noninferiority contrast, $1.05L_{\text{ridge}} - L_{\text{KAHM}}$, is 0.0416102, so that criterion is established. In total, **two of six primary criteria are established**. The overall prespecified criterion is not met.

The raw-RBF comparison illustrates why a point gain is insufficient: its 3.92% estimated gain exceeds the nominal 3% target, but its simultaneous lower margin bound is below zero. That is an unresolved margin claim, not evidence that the procedures are equal. Boosting and the C-selected non-KAHM procedure have lower aggregate point loss; the displayed tests do not themselves certify reverse superiority.

What was actually matched

The bounded KAHM and RBF departure controls share reference geometry, kernels, gates, feature dimensions, transformation rules, and decoder grids. Each has 42 common plus 168 departure coordinates in the declared configuration. Their fitted coefficients, fitted scales, and selected settings remain procedure-specific. Raw RBF uses the 12-coordinate history with at most 64 Nyström coordinates. Boosting and the selected policy have different representations and search spaces. Those comparisons assess complete procedures; they do not isolate one feature block at equal computation.

The original paired Student-t calculation uses 23 degrees of freedom and critical value approximately 2.582017 for a one-sided family allocation of $0.05/6$. These are nominal approximate intervals under the declared sampling design. Arithmetic replay does not validate exact coverage, expand the set of application domains, or turn the research margins into deployment requirements.

5. Verification performed for this revision

Check	Result and scope
Independent mathematical diagnostics	240 normalized-weight queries across four banks; three equal-span examples including redundant coordinates; the two-reference and irregular-reference algebra examples; 480 radial-map pairs; and 36 diagnostic ridge fits, including a single-observation fit and negative, zero, and positive correction scales. All passed the declared 2×10^{-9} tolerance. These small fits are formula diagnostics, not forecasting experiments.
Author-supplied compact verifier	Successfully rerun during recovery of the original numerical supplement. Checks 399 manifest-listed files, 1,920 forecast arrays with 9,584,640 values, eight analysis tables, and all six primary comparisons. The recorded verifier reports 115,024 table cells checked.
Independent compact replay	Newly written code, importing no supplied verifier, checks the same packaged file hashes, all forecast arrays, A-target normalization, the roster, 192 selected-policy identities, and six primary contrasts. Maximum discrepancy among its recomputed primary numeric fields is about 1.263×10^{-14}.
Independent expanded replay	Checks 6,912 input-array hashes, 768 episode-partition/window constructions, 96 paired-noise task pairs, 576 A-normalization values, eight statistical tables, and the stability table. Reconstructs 5,760 horizon rows and 1,920 task rows before the repetition-level analysis. No study fitting or analysis module is imported.
Recorded selection, bound, and resource arithmetic	The expanded audit checks 4,608 B-selection and 4,608 C-selection horizon records, 10,560 recorded selection numeric values, 5,184 bound calculations, and 7,680 resource-accounting values. These are checks of stored scores and formulas, not refitting, feature regeneration, or new timings.
Source and document checks	All 106 files in the supplied source archive match the corresponding Git blobs at the pinned public experiment revision. All 65 original generated assets remain byte-identical after editing. The main and supplement compile without unresolved references/citations, duplicate labels, or overfull boxes.

Useful diagnostic details

The normalized-weight diagnostics include 156 query rows with negative weights, so they do not silently substitute positive convex weights. Their largest affine-target identity discrepancy is approximately 3.55×10^{-15}. The independent ridge solver formulations differ by at most 4.93×10^{-13}; an attaining example reaches both reported bounds. The recorded experimental maximum-error ratio is 1.5534154, and no recorded feature, coefficient, correction, or stability violation was found under the original checks.

Limits of this verification. The principal forecast arrays are original archived data, not new observations. No forecasting models were refitted, no selected fitted objects were loaded, and no hardware timings were repeated. The larger supporting studies were reviewed in the manuscript but were not all independently replayed. Floating-point diagnostics supplement the proof review; they are neither formal proofs nor interval-arithmetic certificates.

6. Recovered evidence and reproduction routes

The missing numerical supplement is now available

The supplied manuscript ZIP contained sources but omitted the numeric supplement referenced in its reproducibility statement. The original numeric ZIP was recovered from the public archived evidence. Its bytes exactly match the checksum in the original release record:

`KAHM_TMLR_Saved_Forecasts.zip`
80,805,506 bytes
`d149ae648b0b380eccf108d8d217566be173182a0ac047982afad4836fb92edd`

Recovery checked all 77 transport parts and their combined archive, then selectively restored numeric inputs, predictions, and summaries. No model-state pickle was extracted or loaded. The public packaging program reproduced the original numeric ZIP exactly and ran its standalone verifier successfully [8].

Separate manuscript and experiment identities

The public experiment and original manuscript source are pinned to repository commit:

`7dda356138675934aab5e327ec667bde0a5eda79`

All 106 supplied source files match their public counterparts at that commit. The present revised article is a **local 31 October 2026 revision**; its edits are not represented by that older public commit. The source patch, current file manifest, and archived original manifest preserve the distinction. No repository changes were published.

Three different operations

1. **Compact numerical replay.** The separate 80.8 MB ZIP contains targets, forecasts, normalization inputs, reported tables, and the author's verifier. It can reproduce the principal forecast statistics offline once dependencies are installed. The source archive also contains a new independent compact contrast-replay program.
2. **Expanded arithmetic audit.** The independent full-evidence program additionally needs original A/B/C/D prepared inputs and per-task fitting summaries. Its episode/window, recorded selection, bound, and resource checks cannot be rerun from the compact ZIP alone. Pinned retrieval and selective restoration commands are included.
3. **Model fitting or selected-state replay.** These are separate operations described in the public repository. Neither was performed for this revision. Agreement of saved-array arithmetic does not imply independent end-to-end experimental replication.

The audit README gives commands for each route. The large transport and full restored inputs are not duplicated inside the source ZIP.

Different prospective records in different studies

The principal experiment's prospective record is local. The earlier direct-reconstruction study has a `PUBLICATION.json` record naming public checkpoint `39174d06dffc`. Its recorded public verification precedes execution by nine seconds; protocol and freeze hashes match. This supports the archived chronology, not an external-registration claim. Both studies' distinct provenance descriptions are retained.

7. Positioning the papers and choosing the next research step

	Risk-certificate manuscript	KAHM forecasting manuscript
Main question	Can a declared representation/decoder class meet a risk tolerance, and can cheaper declared classes be excluded?	When do KAHM departures add raw functions beyond matched coordinates, and does the resulting fitted procedure improve forecasting?
Core result	Calibrated lower and upper risk evidence; a hierarchy of representation assessments under stated assumptions.	Raw-span equalities and a distinctness example; an intercept-inclusive residual bound; matched and complete-procedure comparisons.
Meaning of a guarantee	A statement about a specified class and evaluation law under a simultaneous probability event.	An algebraic statement or deterministic fitted-correction inequality, plus an approximate statistical comparison under the experiment.
Link to adoption	Can support a declared feasibility/cost decision when informative certificates are available.	Provides a candidate feature construction and evidence about where its value is or is not established.

The risk paper's learned implementation primarily uses simplex-valued affinities with constrained decoders. The present paper uses gated reconstruction departures, separate radial transformations, and fitted residual heads. Its baseline and targets are not uniformly bounded. The risk paper's guarantees therefore do not automatically apply to this pipeline. Any future connection must specify the representation, decoder budget, target assumptions, and evaluation law explicitly.

Publication assessment

The present contribution can be argued as a theoretical and empirical study of a representation's behavior. TMLR's stated criteria emphasize well-supported claims and useful, clear findings; a new state-of-the-art score or a novel studied method is not required [6]. This supports the chosen positioning, not a prediction of acceptance.

Before submission, the authors should review the new attribution and proof repair and check exact overlap with related manuscripts. TMLR permits overlap with preprints but restricts reuse from published, accepted, or parallel archival submissions [7]. Distinct research questions alone do not establish compliance; a complete cross-manuscript originality audit was not performed here.

Both PDFs retain anonymous TMLR review typography. The template's "under review" header is a formatting mode, not a statement that a submission has occurred. The editable source ZIP retains author licenses, provenance, and this author-facing review; it is not an anonymous review upload. Nothing has been submitted or published during this work.

Step 3 should test a concrete adoption decision

The next experiment should ask when a user should choose KAHM for specified series, horizons, sample sizes, and resource limits. Predeclare that decision, retain strong comparators and selection costs, and test it on new held-out tasks. The present simulations do not validate a new-domain selection rule. That adoption evaluation remains Step 3.

8. Files, evidence, and references

How to use the delivered files

Start with the revised main PDF and this review. The supporting-studies PDF retains the full auxiliary methods and results. The editable `KAHM_Reconstruction_Prediction.zip` contains `main.tex`, `supplement.tex`, their PDFs, all assets and styles, and a `revision_review/` directory containing this report, the change patch, audit programs and outputs, and provenance records.

The numeric archive `KAHM_TMLR_Saved_Forecasts.zip` is a separate download. It is the original recovered evidence, not a newly generated experiment. The build README explains compilation; the audit README explains the compact and expanded replay routes. All revision checks are dated 31 October 2026. Bibliographic claims below were checked against primary sources; numerical statements in this review are supported by the packaged audit records [8].

References and primary evidence

[1] Deparis, S., Forti, D., and Quarteroni, A. (2014). *A Rescaled Localized Radial Basis Function Interpolation on Non-Cartesian and Nonconforming Grids*. SIAM Journal on Scientific Computing, 36(6), A2745–A2762. doi:10.1137/130947179.

[2] De Marchi, S., Idda, A., and Santin, G. (2017). *A Rescaled Method for RBF Approximation*. In Fasshauer, G. E. and Schumaker, L. L. (eds.), Approximation Theory XV: San Antonio 2016, 39–59. Springer. doi:10.1007/978-3-319-59912-0_3. Author-hosted full text, especially the quotient construction in Section 3.

[3] Kumar, M., Moser, B. A., and Fischer, L. (2024). *On Mitigating the Utility-Loss in Differentially Private Learning: A New Perspective by a Geometrically Inspired Kernel Approach*. Journal of Artificial Intelligence Research, 79, 515–567. doi:10.1613/jair.1.15071.

[4] Kumar, M., and Kargaran, S. (2026). *Prediction Limits and Koopman Closure of Geometry-Induced Soft State Abstractions*. arXiv:2609.32652v3. Version 3 full text.

[5] Kumar, M., and Kargaran, S. (2026). *A Hierarchy of Risk Certificates for Forecasting Representations*. User-supplied unpublished manuscript; revised in Step 1. No journal acceptance or public identifier is asserted for this manuscript.

[6] Transactions on Machine Learning Research. Acceptance Criteria. Accessed 31 October 2026.

[7] Transactions on Machine Learning Research. Submission Guidelines and Editorial Policies and Author Guidelines. Accessed 31 October 2026.

[8] Original experimental evidence in the pinned reconstruction-mechanism directory; original numeric-release manifest and recovered `PACKAGE_SUMMARY.json`; independent revision outputs `method_theory_checks.json`, `independent_compact_forecast_checks.json`, and `independent_saved_forecast_checks.json`; source-provenance, restoration, and final-build records. These are supplied in the revised source archive's `revision_review/` directory.